

Status	Finished
Started	Sunday, 12 January 2025, 5:29 PM
Completed	Sunday, 12 January 2025, 5:46 PM
Duration	16 mins 54 secs

Question **1**

Correct

Marked out of 1.00

 Flag question

You are a bank account hacker. Initially you have 1 rupee in your account, and you want exactly **N** rupees in your account. You wrote two hacks, first hack can multiply the amount of money you own by 10, while the second can multiply it by 20. These hacks can be used any number of time. Can you achieve the desired amount **N** using these hacks.

Constraints:

$1 \leq T \leq 100$
 $1 \leq N \leq 10^{12}$

Input

- The test case contains a single integer N.

Output

For each test case, print a single line containing the string "1" if you can make exactly N rupees or "0" otherwise.

SAMPLE INPUT

1

SAMPLE OUTPUT

1

SAMPLE INPUT

2

SAMPLE OUTPUT

0

Answer: (penalty regime: 0 %)

Reset answer

```
1 int myFunc(int n) {  
2     // Check divisibility by 10 or 20, or if n is 1  
3     if ((n % 10 == 0) || (n % 20 == 0) || (n == 1)) {  
4         return 1;  
5     }  
6     return 0; // Otherwise, return 0  
7 }
```

	Test	Expected	Got	
✓	<code>printf("%d", myFunc(1))</code>	1	1	✓
✓	<code>printf("%d", myFunc(2))</code>	0	0	✓
✓	<code>printf("%d", myFunc(10))</code>	1	1	✓
✓	<code>printf("%d", myFunc(25))</code>	0	0	✓
✓	<code>printf("%d", myFunc(200))</code>	1	1	✓

Passed all tests! ✓

Question **2**

Correct

Marked out of 1.00

Flag question

Find the number of ways that a given integer, X , can be expressed as the sum of the N^{th} powers of unique, natural numbers.

For example, if $X = 13$ and $N = 2$, we have to find all combinations of unique squares adding up to 13 . The only solution is $2^2 + 3^2$.

Function Description

Complete the `powerSum` function in the editor below. It should return an integer that represents the number of possible combinations.

`powerSum` has the following parameter(s):

X : the integer to sum to

N : the integer power to raise numbers to

Input Format

The first line contains an integer X .

The second line contains an integer N .

Constraints

$$1 \leq X \leq 1000$$

$$2 \leq N \leq 10$$

Output Format

Output a single integer, the number of possible combinations calculated.

Sample Input 0

10

2

Sample Output 0

1

Explanation 0

If $X = 10$ and $N = 2$, we need to find the number of ways that 10 can be represented as the sum of squares of unique numbers.

$$10 = 1^2 + 3^2$$

This is the only way in which 10 can be expressed as the sum of unique squares.

Sample Input 1

100

2

Sample Output 1

3

Explanation 1

$$100 = (10^2) = (6^2 + 8^2) = (1^2 + 3^2 + 4^2 + 5^2 + 7^2)$$

Sample Input 2

100

3

Sample Output 2

1

Explanation 2

100 can be expressed as the sum of the cubes of **1, 2, 3, 4**.

(1 + 8 + 27 + 64 = 100). There is no other way to express **100** as the sum of cubes.

Answer: (penalty regime: 0 %)

Reset answer

```

1 #include <math.h>
2
3 int powerSum(int x, int m, int n) {
4     if (x == 0) {
5         return 1; // Base case: when the sum is achieved
6     }
7
8     if (x < 0) {
9         return 0; // Base case: invalid partition
10    }
11
12    int c = 0;
13
14    // Loop through possible values
15    for (int i = m; pow(i, n) <= x; i++) {
16        int p = pow(i, n); // Calculate i^n
17        c += powerSum(x - p, i + 1, n); // Recursive call with reduced sum
18    }
19
20    return c; // Return the total count
21 }

```

	Test	Expected	Got	
✓	printf("%d", powerSum(10, 1, 2))	1	1	✓

Passed all tests! ✓